Laxmi Charitable Trust's

Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce Department of Information Technology

(B.Sc.I.T Semester V)

Full Stack development MERN Practical

Practical– VIII

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Date of Assignment: 12-08-26	Date/Time of Submission: 12-08-26

Practical 8: API Creation and Integration

Aim:
To create and integrate a REST API using Express, MongoDB and React.

A. Create a Simple REST API using Express

Steps:

1. Create `APIPractical` folder and open it in VS Code.
2. Run `npm init -y`.
3. Install `mongoose`, `express` and `cors`.
4. Create `server.js`.
5. Connect Express with MongoDB `CollegeDB`.
6. Create a simple GET route `/`.
7. Run `node server.js`.
8. Open http://localhost:3000.

server.js:-

```
const express = require("express");

const mongoose = require("mongoose");

const cors = require("cors");



const app = express();

const PORT = 3000;
```

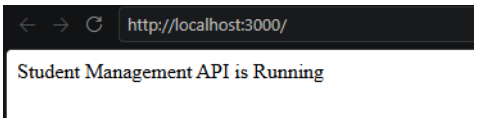
```
// Middleware
app.use(cors());
app.use(express.json());


// Connect MongoDB
mongoose.connect("mongodb://127.0.0.1:27017/CollegeDB")
  .then(() => {
    console.log("MongoDB Connected Successfully");
  })
  .catch((error) => {
    console.log("MongoDB Connection Failed");
    console.log(error);
  });


// Simple REST API
app.get("/", (req, res) => {
  res.send("Student Management API is Running");
});


app.listen(PORT, () => {
  console.log(`Server is running on http://localhost:${PORT}`);
});
```

Output:



B. Implement GET API

Steps:

1. Create `CollegeDB` database and `students` collection in MongoDB.
2. Insert student records.
3. Add `GET /students` API in `server.js`.
4. Run the server.
5. Open http://localhost:3000/students.

Server.js:-

```
const express = require("express");

const mongoose = require("mongoose");

const cors = require("cors");



const app = express();

const PORT = 3000;



// Middleware

app.use(cors());

app.use(express.json());



// Connect MongoDB

mongoose.connect("mongodb://127.0.0.1:27017/CollegeDB")

  .then(() => {

    console.log("MongoDB Connected Successfully");

  })

  .catch((error) => {

    console.log("MongoDB Connection Failed");

    console.log(error);

  });
```

```
// Simple REST API

app.get("/", (req, res) => {

  res.send("Student Management API is Running");

});

app.get("/students", async (req, res) => {

  try {

    const students = await mongoose.connection.db

      .collection("students")

      .find()

      .toArray();



    res.json(students);

  } catch (error) {

    res.status(500).json({

      message: "Error fetching students"

    });

  }

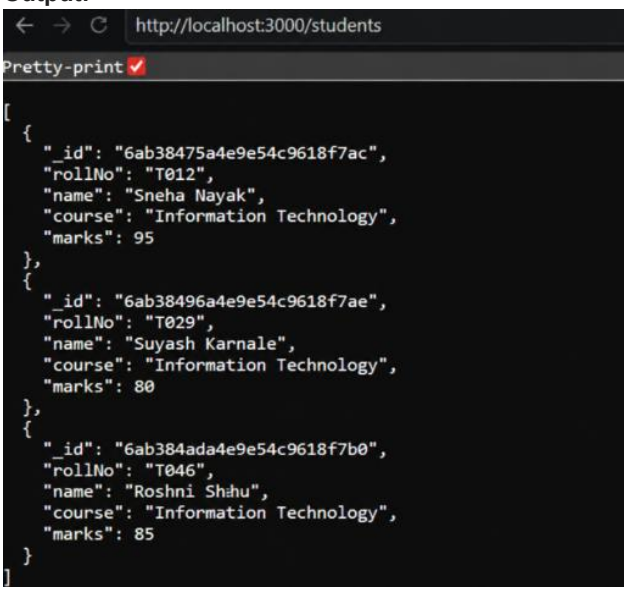
});

app.listen(PORT, () => {

  console.log(`Server is running on http://localhost:${PORT}`);

});
```

Output:



C. Fetch API Data in React

Steps:

1. Create React project using Vite.
2. Install Axios.
3. Add API request in `App.jsx`.
4. Start Express server using `node server.js`.
5. Start React using `npm run dev`.
6. Open the React URL in the browser.

Code:-

```
import { useEffect, useState } from "react";

import axios from "axios";

function App() {

const [students, setStudents] = useState([]);

useEffect(() => { axios.get("http://localhost:3000/students")

.then((response) => {setStudents(response.data);})



.catch((error) => {console.log(error);}); }, []);

return (

<div>

<h1>Student List</h1>
```

```
<table border="1">

<thead>

<tr>

<th>Roll No</th>

<th>Name</th>

<th>Course</th>

<th>Marks</th>

</tr>

</thead>

<tbody>

{students.map((student) => (

<tr key={student._id}>

<td>{student.rollNo}</td>

<td>{student.name}</td>

<td>{student.course}</td>

<td>{student.marks}</td>

</tr>

))}

</tbody>

</table>

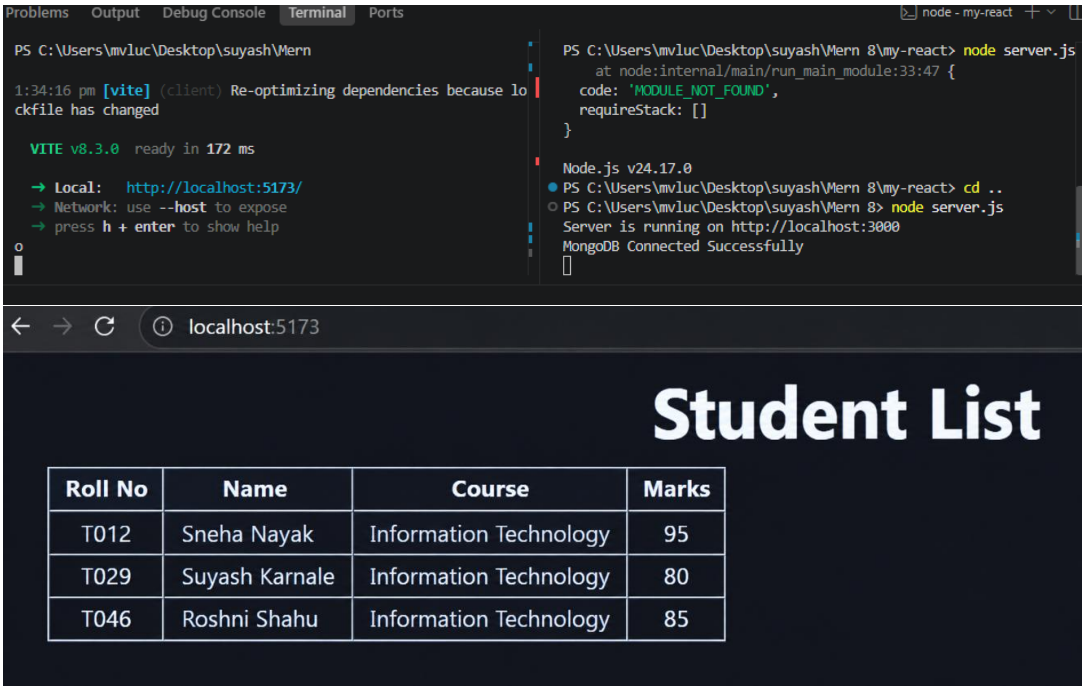
</div>

);

}

export default App;
```

Output:



D. Send Data using POST Request

Steps:

1. Add POST `/students` API in `server.js`.
2. Create student input form in React.
3. Enter Roll No, Name, Course and Marks.
4. Click **Add Student**.
5. Axios sends the data to Express.
6. Express stores the data in MongoDB.
7. Student list is refreshed.

Code:-

```
const express = require("express");

const mongoose = require("mongoose");

const cors = require("cors");



const app = express();

const PORT = 3000;



// Middleware
```

```
app.use(cors());

app.use(express.json());



// Connect MongoDB

mongoose.connect("mongodb://127.0.0.1:27017/CollegeDB")

  .then(() => {

    console.log("MongoDB Connected Successfully");

  })

  .catch((error) => {

    console.log("MongoDB Connection Failed");

    console.log(error);

  });



// Simple REST API

app.get("/", (req, res) => {

  res.send("Student Management API is Running");

});

app.get("/students", async (req, res) => {

  try {

    const students = await mongoose.connection.db

      .collection("students")

      .find()

      .toArray();



    res.json(students);

  } catch (error) {

    res.status(500).json({
```

```
      message: "Error fetching students"
    });
  }
});
app.post("/students", async (req, res) => {
  try {
    const student = req.body;

    await mongoose.connection.db
      .collection("students")
      .insertOne(student);

    res.json({
      message: "Student added successfully",
      student: student
    });
  } catch (error) {
    res.status(500).json({
      message: "Error adding student"
    });
  }
});
app.listen(PORT, () => {
  console.log(`Server is running on http://localhost:${PORT}`);
});
```

App.jsx:-

```
import { useEffect, useState } from "react";

import axios from "axios";

function App() {

 const [students, setStudents] = useState([]);

 const [rollNo, setRollNo] = useState("");

 const [name, setName] = useState("");

 const [course, setCourse] = useState("");

 const [marks, setMarks] = useState("");

 // Get students

 const getStudents = () => {axios.get("http://localhost:3000/students")
.then((response) => {setStudents(response.data);})

 .catch((error) => {console.log(error);});

 };

 useEffect(() => {getStudents();}, []);

 // Add student

 const addStudent = () => {const newStudent = {  rollNo: Number(rollNo),

 name: name,

 course: course,

 marks: Number(marks)

 };

 axios.post("http://localhost:3000/students", newStudent)  .then((response) =>
{alert(response.data.message);  setRollNo("");

 setName("");

 setCourse("");

 setMarks("");

 getStudents();

 })
```

```
.catch((error) => {

console.log(error);

});

};

return (

<div>

<h1>Student Management System</h1>

<h2>Add Student</h2>

<input

type="number"

placeholder="Roll No"

value={rollNo}

onChange={(e) => setRollNo(e.target.value)}  />

<br /><br />

<input

type="text"

placeholder="Student Name"

value={name}

onChange={(e) => setName(e.target.value)}  />

<br /><br />

<input

type="text"

placeholder="Course"

value={course}

onChange={(e) => setCourse(e.target.value)}  />

<br /><br />

<input
```

```
type="number"

placeholder="Marks"

value={marks}

onChange={(e) => setMarks(e.target.value)}  />

<br /><br />

<button onClick={addStudent}>

Add Student

</button>

<hr />

<h2>Student List</h2>

<table border="1">

<thead>

<tr>

<th>Roll No</th>

<th>Name</th>

<th>Course</th>

<th>Marks</th>

</tr>

</thead>

<tbody>

{students.map((student) => (

<tr key={student._id}>

<td>{student.rollNo}</td>  <td>{student.name}</td>

<td>{student.course}</td>  <td>{student.marks}</td>

</tr>

))}

</tbody>
```

```
 </table>

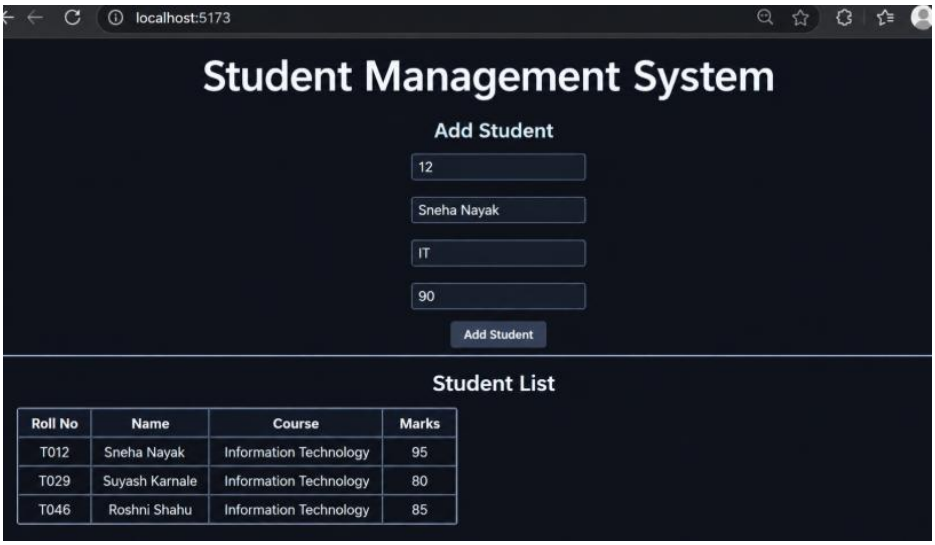
 </div>

 );

}

export default App;
```

Output:



Result

The REST API was successfully created using Express and MongoDB, and integrated with a React application using GET and POST requests.